

Homework 5

40 points

1. Find the rank of the following matrices **2*5 = 10 points**

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & 3 \\ 2 & 8 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

2. What is the maximum possible rank of a 4×6 matrix? **2 points**

3. For what value of k does the following matrix have rank 2? **3 points**

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 2 & k \end{bmatrix}$$

4. If A is a 2×3 matrix, what is the order (dimensions) of A^T ? **2 points**

5. Is the following matrix an orthogonal matrix? Show your steps **3 points**

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

6. What is the determinant of a square matrix with 2 identical rows? **2 points**

7. For what value of k will the determinant of the given matrix be 5? **3 points**

$$\begin{bmatrix} k & 1 \\ 3 & 2 \end{bmatrix}$$

8. Find the determinant of: **5 points**

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 3 & 4 & 0 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

Use co-factor expansion and mention the row/column used

9. Find the inverse of the following matrix: **6 points**

$$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

10. For what value of k is the given matrix invertible? **4 points**

$$\begin{bmatrix} 1 & 2 \\ 3 & k \end{bmatrix}$$

Explain your answer